

APPLICATION

CDMetaPOP: an individual-based, eco-evolutionary model for spatially explicit simulation of landscape demogenetics

Erin L. Landguth^{1*}, Andrew Bearlin², Casey C. Day³ and Jason Dunham⁴

¹Division of Biological Sciences, University of Montana, 32 Campus Drive, Missoula, MT 59846, USA; ²Environmental Affairs Division, Seattle City Light, 700 5th Avenue, Seattle, WA 98124, USA; ³Department of Forestry and Natural Resources, Purdue University, 195 Marsteller Street, West Lafayette, IN 47909, USA; and ⁴Forest and Rangeland Ecosystem Science Center, U.S. Geological Survey, 3200 SW Jefferson Way, Corvallis, OR 97330, USA

Summary

1. Combining landscape demographic and genetics models offers powerful methods for addressing questions for eco-evolutionary applications.
2. Using two illustrative examples, we present Cost–Distance Meta-POPulation, a program to simulate changes in neutral and/or selection-driven genotypes through time as a function of individual-based movement, complex spatial population dynamics, and multiple and changing landscape drivers.
3. Cost–Distance Meta-POPulation provides a novel tool for questions in landscape genetics by incorporating population viability analysis, while linking directly to conservation applications.

Key-words: connectivity, dispersal, dynamic landscapes, gene flow, genotype–environment associations, growth rate, harvesting, landscape resistance, sampling, stream networks

Introduction

Landscape genetics has focused on assessing the degree to which landscapes between populations facilitate or impede dispersal movement and gene flow (Holderegger & Wagner 2008). Classic metapopulation ecology focuses on the effects of local environmental conditions on population dynamics (i.e. extinction and recolonization), which can greatly influence genetic differentiation and diversity in study systems (McCauley 1991; Banks *et al.* 2013). Therefore, the importance of additionally considering demographic responses with local environmental conditions in landscape genetics studies has become more apparent (e.g. Frank, Piccolo & Baret 2011; Pflüger & Balkenhol 2014).

Thus, we have created Cost–Distance Meta-POPulation (CDMetaPOP), a spatially explicit, individual-based, forward-in-time simulator for landscape demographic and genetics (defined herein as landscape demogenetics). CDMetaPOP was specifically developed for species living in dynamic riverine landscapes ('riverscapes'; Ward 1998; Fausch *et al.* 2002). However, the model is suitable for a spectrum of processes that operate in any heterogeneous environment (e.g. terrestrial; Landguth, Holden & Mahalovich 2015; or aquatic; Schlosser 1995; Rieman & Dunham 2000) where spatial arrangement of habitats, habitat-specific demographic variables and behavioural variables describing the dispersal

characteristics of the species can be defined (Pulliam, Dunning & Liu 1992). In this sense, CDMetaPOP can simulate a range of spatially explicit processes that describe, and have given rise to, generalized metapopulation theory (Hanski & Simberloff 1997). CDMetaPOP can accommodate simulations involving up to hundreds of thousands of individuals and supports prediction of landscape demogenetic responses to variable model inputs or simulated scenarios. CDMetaPOP's spatially explicit representation of functional connectivity (e.g. Manel & Holderegger 2013), local environmental conditions, population demography and genetic processes provide a powerful framework for investigating a host of processes that operate in landscapes.

Here, we describe the major features of the program (e.g. selection-driven processes, growth–environment associations). We show two examples that illustrate applications of the model: (i) influences of spatial processes on the ability to control a non-native, invasive species in a stream network and (ii) sampling design and effort around insertion and removal of barriers. Lastly, we discuss the specific advantages and opportunities for applied simulations that CDMetaPOP provides as compared to foundational forward-in-time landscape genetics simulators.

Main processes

Cost–Distance Meta-POPulation simulates demogenetic processes as interactions between individuals located in a number

*Correspondence author. E-mail: erin.landguth@mso.umt.edu

of ‘patches’. Our use of the term ‘patch’ is flexible and may differ from other definitions used in ecology (e.g. Dunham, Rie- man & Peterson 2002). In the world of CDMetaPOP, patches are the fundamental units of a landscape within which individuals share a common environment. Landscape heterogeneity, in turn, is defined by spatially explicit parameterization of environmental variables among patches. Movement of individuals among patches is a function of any number of spatially explicit resistance (or permeability) surfaces. Within patches, individuals undergo growth, reproduction, migration and mortality, and the resulting genetic processes are simulated over a user-specified given time step at the individual level. The conceptual life cycle with a detailed breakdown of each specific process can be found in the user manual. Note that each of the modules described below can be essentially turned on or off depending on the objective of the simulation study.

DYNAMIC LANDSCAPES

Cost–Distance Meta-POPulation supports the ability to alter the simulation environment and parameter space through time. Simulations using this functionality can address a variety of questions about dynamic factors influencing landscape demogenetic processes (e.g. Landguth *et al.* 2010; Bocedi *et al.* 2014). For example:

- How does spatial and/or temporal variation in temperature influence growth and/or movement capability of individuals?
- How will insertion or removal of barriers influence demogenetic connectivity?
- How will habitat improvement or harvesting actions influence population and genetic viability?

These examples can be addressed separately or in combination by CDMetaPOP. Scenario-driven questions are commonplace in applied and basic research, but the capability of CDMetaPOP to account for these processes in a dynamic and spatially explicit context can provide rapid insights into landscape processes that can be extremely difficult and expensive to study or monitor empirically.

NEUTRAL AND SELECTION-DRIVEN GENETICS

Cost–Distance Meta-POPulation builds on the architecture of CDPOP (Landguth & Cushman 2010) for modelling neutral genetics with a number of significant additions. The user starts by specifying an unlimited number of loci and alleles to simulate. Individual genotypes can then be initialized randomly, by allele frequency files that can vary spatially, or by empirical genotypes. Offspring then receive genotypes based on Mendelian inheritance. There are a number of mutational models that can be considered (*k*-allele and variations on stepwise) with rate chosen by the user. An optional maternal-DNA marker can be used, as well as the option to simulate single nucleotide polymorphisms.

Cost–Distance Meta-POPulation implements natural selection analogously to the fitness landscape of allele frequencies as originally envisioned by Wright (1932). This functionality

enables extension of landscape genetic analyses to explicitly investigate the links between gene flow and selection at the level of the individual (see Landguth, Cushman & Johnson 2012a; Landguth *et al.* 2012b). Implicit in this process is the need to define spatially explicit fitness values (i.e. selection coefficients) for each genotype under selection (i.e. one- or two-locus selection models; see Hartl & Clark 2007) that either replace or modify demographic parameters for individuals in the patch. For example, relative spatial fitness values must be specified for the nine genotypes of the two-locus model (i.e. *AABB*, *AaBB*, ..., *Aabb*, *aabb*) at each patch. Selection is implemented through differential survival as a function of the relative fitness at the patch location where the individual settles with options for offspring or adult differential survival. For example, the selection module can be applied to model survival of individuals in warming streams and examine how quickly alleles for warmer thermal tolerance ‘invade’ these areas.

In addition, selection for genes governing maturation, growth and migration mechanisms can be simulated. For example, earlier-maturing individuals could be associated with a genotype with faster growth (Sloat *et al.* 2014), with attendant consequences for survival and reproduction (e.g. size-selective harvest, predation risk; McClure *et al.* 2008). In addition, simulations can ask how alleles for earlier migration tied to changes in thermal and hydrologic regimes alter demogenetics in headwater streams (e.g. Carlson & Seamons 2008; Anderson & Beer 2009; Kovach, Gharrett & Tallmon 2012).

DEMOGRAPHICS

Cost–Distance Meta-POPulation allows the emergence of complex spatial demographic stochasticity through individual responses to environmental and population variables that may be specified at the level of the individual (or summarized for a set of individuals) and implemented as Markov chain Monte Carlo variables. It is important to distinguish that the membership of these classes is dynamic and hysteretic, unlike as would be typical of state variable models (Caswell 1989). The following sections describe how maturation, fecundity, mortality and density dependence operate within the program, keeping in mind that most references herein are specific to fisheries applications, but can be adapted to consider terrestrial systems, as well.

POPULATION STRUCTURE

Cost–Distance Meta-POPulation uses a powerful hybrid approach to population modelling that blends individual-based processes with class-based demographic (age/stage) controls, genetic-environment-based fitness surfaces and patch-based landscape attributes. This allows spatial population structure and dynamics to be explicitly modelled with unconstrained complexity as the user requires. For example, probabilistic parameters, such as individual growth rate, fecundity or survivorship, can be defined as functions of any combination of patch-, class-, genetic- or population-based variables.

STOCHASTICITY

Cost–Distance Meta-POPulation uses probabilistic rates (e.g. birth) and calculates binomial outcomes for individuals, meaning that demographic stochasticity is implicit in the model. Environmental stochasticity can be included by specifying mean and variance for parameters of interest. Correlation structures between these variables can be specified using a correlation matrix. This functionality allows extremely flexible specification of sources of random environmental stochasticity and correlation/covariance structures. For example, variation in egg mortality might be expected to be negatively correlated to environmental variation (i.e. when conditions are good, mortality should decline).

MATURATION

Maturation is specified by the user as a function of class variables modified by patch and/or population variables as required, with two options provided. If the user specifies age-controlled parameters (e.g. many endotherms), then the probability of becoming a mature, reproducing individual at a given sex and age is provided by the user. If size control is specified (e.g. many ectotherms), then the probability of maturation for individuals is modelled using a logistic function of size, where parameters can be fit based on empirical data (e.g. as a function of sex and length; Downs, White & Shepard 1997). In the examples presented below, maturation is controlled as a function of size, which is in turn dependent on the temperature experienced by the individual at their located patch; thus, maturation probability is an integration of spatial heterogeneity of temperature, growth and movement.

FECUNDITY

Cost–Distance Meta-POPulation offers the option to specify $F \sim P(\lambda)$ where λ is the mean number of offspring per mature female at a given size or age class. The user may specify F as a function of a Poisson or Gaussian distribution with appropriate consideration of the implicit assumptions (e.g. truncations or other continuity corrections; Todd & Ng 2001), as well as a constant draw to validate models under Wright–Fisher assumptions (see user manual). If age control is specified, then a mean and standard deviation for offspring per mature female at a given age are provided by the user. If size control is specified, then the mean offspring is derived as a function of length with three functional forms available: (i) linear (e.g. Downs, White & Shepard 1997), (ii) exponential (e.g. Bowerman 2013) and (iii) power (e.g. Schill *et al.* 2010). The user may specify the parameters of these functions using size–fecundity data, or by using literature values. Reproductive frequency and structure (e.g. sexual monogamy or hermaphroditic with selfing), iteroparity and semelparity (or skipped spawning) can also be considered for all individuals (e.g. Rideout & Tomkiewicz 2011).

MORTALITY

Several options are available to regulate mortality. Mortality variables can be specified to operate by age class, size class and/or patch. At each time step, mortalities can be compounded to simulate different rates of additional mortality operating on individuals (or groups of individuals, e.g., populations, size or age classes). In the examples below, we demonstrate mortality due to anthropogenic (e.g. harvest pressure; see Example #1) and environmental impacts (e.g. flow or temperature; Ray, Peacock & Dunham 2007; Peacock & Dochtermann 2012). In addition, mortality in these examples is implemented at two discrete time steps (e.g. intra-annual) in the model. The first represents mortality applied to individuals of a given class when they return to their natal grounds or stray into new natal grounds; the second represents mortality applied to individuals that are overwintering. Finally, the user may specify spatially explicit mortality applied to the litter class or group of eggs at birth location.

DENSITY-DEPENDENT MORTALITY

Competition for limited resources may be implemented via a simple ‘ceiling’ function (i.e. population size will grow until patch carrying capacity (K) is reached and then a random removal of individuals per patch occurs; e.g., Balloux 2001) or via a density-dependent function with a flexible hierarchical specification of patch- and class-structured carrying capacities (e.g. Ray & Hastings 1996). This form of density dependence was chosen to allow incorporation of individual differences in the ability to compete for space in a patch (e.g. on the basis of size) and was heavily influenced by the choice of the example subject (salmonid fishes). The mechanism used here also has the advantage of not relying on population-based controls, as might be required if density dependence were based on recruitment limitation (e.g. Atlas *et al.* 2015).

The mechanism of density dependence implemented here is based on a set of logistic equations with the ability to specify an ideal distribution of K available to each of the specified classes (age or size cohort) within a given patch (see user manual for a complete formulation). In this implementation, the R_0 parameter hierarchically modifies the *a priori* allocation of K available to a class, as a function of both class- and patch membership. Together with recursive reallocation of underutilized capacity according to the relative abundances of each class (or cohort) competing within the patch, this mechanism simulates inter- and intracohort competition, with the user able to specify the degree of overlap and partitioning of habitat patch resources for each class.

GROWTH

Individual size initialization is accomplished by specifying a size (e.g. length) mean and standard deviation for each class in the input files. Size is allocated using a normal draw for each age class. There are four growth options: (i) ‘N’ – turn off growth altogether, (ii) ‘known’ – grow individuals at a known

increment, (iii) 'vonB' – grow individuals based on the von Bertalanffy equation (von Bertalanffy 1938), or (iv) 'temperature' – use spatially explicit temperature values to drive growth for each individual. The latter option requires empirical data that are fit to relationships between length-environmental curves (see user manual for complete derivations and example empirically derived relationships). Growth may be evaluated and applied at two separate timing events within the user-specified time unit (or growing days), for example, simulating when individuals are at their natal grounds or when they are at some other location (e.g. overwintering).

INDIVIDUAL MOVEMENT

Landscape structure governing movement of individuals between patches is incorporated through resistance (or permeability) surfaces whose values represent the stepwise cost from patch-to-patch of crossing each location (Spear, Cushman & McRae 2015). The model simulates movement as probabilistic functions of cumulative cost across these resistance surfaces (Landguth & Cushman 2010). Cost–distance matrix examples include least-cost path, Euclidean or riverine distance and can be generated from any program the user chooses (e.g. PATHMATRIX; Ray 2005; CIRCUITSCAPE; McRae 2006; UNICOR; Landguth *et al.* 2012b; GDISTANCE; van Etten 2015).

Cost–Distance Meta-POPulation considers up to four discrete cost–distance matrices for each sex, controlling emigration from natal patch, homing to natal patch, mating and straying, each with dimensions *patch* × *patch*. Together these matrices control where an individual can move spatially, how they search for mates, and in the case of our salmonid study subjects, how they stray (return to spawn in a patch other than the one they were born in). The user specifies the maximum dispersal distance an individual can travel on the landscape, which can be used for species-specific known movement thresholds or helping to define complete barriers in terms of cost–distances (Landguth *et al.* 2010). There are eight optional movement functions that, together with known movement thresholds, determine the probability of moving to a patch (or selecting a mate partner) as a function of cost–distance (see user manual for more details on program-specific movement functions that convert cost–distances to probability). The user also has the option of supplying the probability matrix directly, which allows for flexibility beyond the program-specific movement functions that depend on cumulative cost.

In addition, asymmetrical cost–distance matrices can be given to test hypotheses of movement in the presence of an environmental gradient (e.g. hilly terrain, wind resistance or one-way vs. two-way barriers; e.g., 'gdistance' vignette; van Etten 2015). If asymmetrical probabilities exist, such as for emigration away from natal patch and homing back to natal patch (i.e. $\text{Prob}(\text{AB}) < \text{Prob}(\text{BA})$ for natal patch A and emigrated patch B), then relative probability differences are calculated and used to determine success of migration back to natal patch. Finally, any of the movement processes (mating, emigration away from natal patch, homing back to natal patch and

straying) can be optionally deactivated to tailor the model towards specific study systems (e.g. aquatic vs. terrestrial).

Example applications

EXAMPLE #1: MANAGED REMOVAL OF NON-NATIVE SPECIES

Control of non-native species in riverscapes is a major conservation and restoration challenge, as a species' demographic resilience, and connectivity within networks can limit the ability of suppression or eradication efforts to influence populations (e.g. Peterson, Fausch & White 2004a). As an example of the ability of CDMetaPOP to simulate complex demographics, we modelled the managed removal of the non-native Eastern Brook Trout (EBT; *Salvelinus fontinalis*) in north-eastern Washington, USA (Fig. 1a) on a spatially heterogeneous landscape. We demonstrate the effects on demographics of three important components of the model: mortality, movement rates and landscape effects. At each time step, a patch-based density-dependent mortality is computed by partitioning space in each patch for each size/age class of fish. In a management scenario, additional mortality associated with mechanical removal of the larger, more catchable individuals, produces an inherent compensatory response to suppression (Ricker 1954; Meyer, Lamansky & Schill 2006; Carmona-Catot *et al.* 2010), as survival will increase when population size is reduced. In addition to density-dependent mortality, an additional mortality can be applied to any combination of patch, age or size class memberships in an additive (simulating processes operating at different points in time) or multiplicative (simulating independent processes operating simultaneously) formulation. For example, specifying 100% patch-based mortality, it is possible to simulate chemical eradication of fish in an entire stream reach. Patch mortality at some rate <100% would simulate incomplete eradication or 'suppression' of non-natives.

The second functionality we demonstrate in this example is movement. Specifically, stray rate, or the rate at which individuals fail to return to their natal patch and instead colonize or disperse to some other patch on the landscape (Quinn 2005). Stray rate is important for maintaining genetic connectivity, but can also play a major role in the recovery and reestablishment of suppressed or eradicated populations (Peterson, Fausch & White 2004a). Finally, we demonstrate the effect of landscape resistance on population demographics through the use of barriers to movement (i.e. dams and culverts).

We modelled the suppression and eradication of EBT in north-eastern Washington in accordance with a management plan developed by Seattle City Light (SCL 2010) (Fig. 1a). The goal of this program is to reduce EBT populations to the extent that native populations of Westslope Cutthroat Trout (WCT; *Oncorhynchus clarkii lewisi*) are able to recover and coexist with the suppressed EBT. We modelled a single cycle of suppression (12 years) to explore rates of population recovery. To simulate efforts to control EBT, we simultaneously implemented two independent and spatially discrete methods requiring different formulation of mortality controls for each. First,

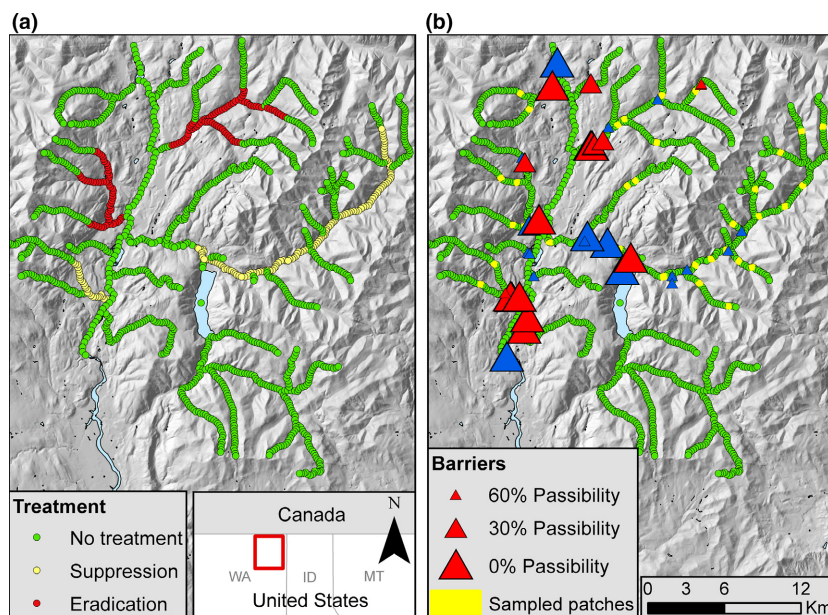


Fig. 1. Simulation set-up for Examples. The examples study area extent with 1384 total patches. (a) Suppression and eradication zones for EBT (Example #1). (b) Example #2's sampled patches (40 yellow patches) for WCT with barrier locations (triangles: blue indicating removed barriers). Barrier locations used in both Example #1 and #2.

eradication was simulated using patch mortality of 100% (to simulate the use of piscicides; Buktenica *et al.* 2013) at discrete patches identified in the treatment plan. Secondly, to simulate suppression of populations by electrofishing (<100% effective; unpublished SCL data; Fig. 1a), we implemented size-based mortality at selected patches (simulating size-dependent capture rates; unpublished SCL data). To demonstrate the effect of treatments, we modelled EBT both with and without the additional harvest mortality described here (Fig. 2). To explore how movement effects recolonization during the (multiyear) period of treatments, we simulated individual stray rates of 0%, 0.1% and 1%, demonstrating how recovery time operates as a function of movement (Fig. 2a). Finally, to demonstrate the role of landscape resistance in population recovery, we simulated landscapes both with and without the existence of barriers placed downstream from eradication zones (Fig. 1b).

EXAMPLE #2: SAMPLING AROUND DETECTION AND REMOVAL OF BARRIERS

Understanding how genetic patterns respond to landscape change and the ability to detect this change with sampling design and effort is an important topic in landscape genetics (see Balkenhol & Fortin 2016). In this example, we illustrate the use of the dynamic landscape module (i.e. systematic variability in landscapes introduced through time) and mimic sampling through capture probabilities (i.e. electrofishing; Chelgren & Dunham 2015).

We considered the same spatial extent and patch delineation as for Example #1 (see Fig. 1b). We parameterized the simulation runs based on a WCT model for the study area (see user manual for how WCT empirical data were used to inform each parameter). We first used a 25-year population dynamics burn-in (no genetics exchange) followed by genetic exchange over a 200-year isolation-by-riverine distance hypothesis. We

then inserted barriers (i.e. dams and culverts) of varying permeability (0–60%; Walker *et al.* 2015) on the riverscape (Fig. 1b triangles) and simulated genetic exchange for 100 years at which time we removed a few key complete barriers (i.e. 0% permeability; Fig. 1b blue triangles) for an additional 100-year simulation. We used the landscape connectivity program, UNICOR (Landguth, Cushman & Johnson 2012a; Landguth *et al.* 2012b), to create asymmetrical pairwise patch-to-patch costs associated with traversing upstream or downstream over barriers. We sampled 40 predetermined patch locations (Fig. 1b yellow points) and specified size-based gear-selectivity and first-pass probability of capture (unpublished SCL technical report; Peterson, Thurow & Guzevich 2004b) for each WCT size class. The program produced 0/1 detection and recapture information for each individual through time. We tracked the inbreeding coefficient (F ; Wright 1922) over the 425-year simulation for the total population (all individuals in the 1384 patches), all individuals in the 40 patches and the captured individuals in the 40 patches (Fig. 3). Samples underestimated the absolute value of the 'true' F , but relative patterns over time are consistent with an approximate 50-year lag time in the sampled response in F when barriers were removed.

Discussion

Simulation modelling studies in landscape genetics have recently begun to proliferate, as individual-based, spatially explicit, genetic modelling tools become more available (e.g. quantiNEMO, Neuenschwander *et al.* 2008; CDPOP, Landguth & Cushman 2010; SimAdapt, Rebaudo *et al.* 2013; MetaPopGen, Andrello & Manel 2015). Landscape genetics studies are increasingly coupling empirical data with simulation designs (e.g. DiLeo *et al.* 2013; Castillo *et al.* 2014). Because essential model features and simplifying assumptions will vary from study to study, it is not surprising that the available

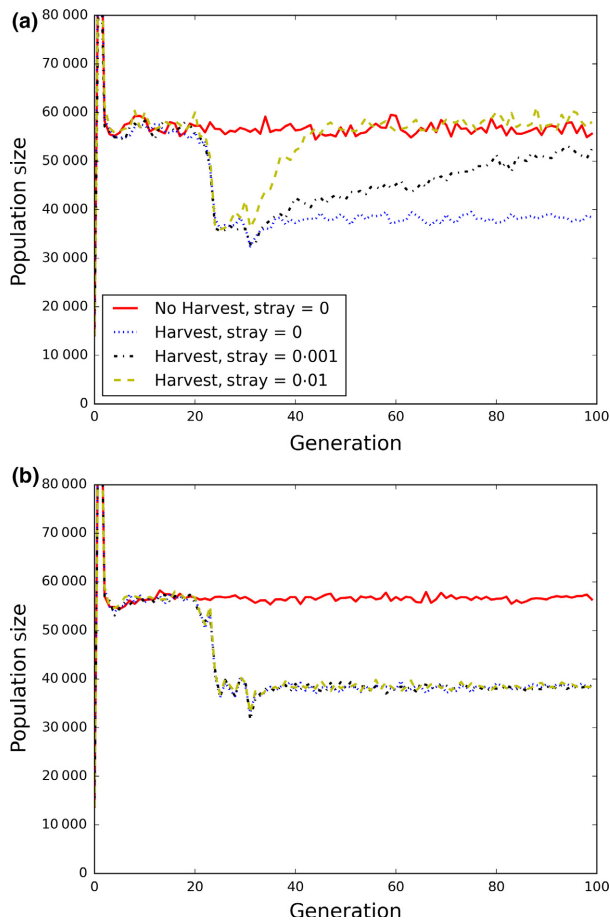


Fig. 2. Example #1 managed removal of non-native species: Population size for Eastern Brook Trout over time across alternate landscapes with (a) barriers removed and (b) barriers included. Three processes are illustrated: (i) The managed suppression and eradication of the invasive species, which shows a decline in population with removal compared with the no removal scenarios (red-solid lines). (ii) The recovery of the population as a function of the proportion of individuals that stray from their natal patch (a). A new lower population equilibrium is observed with no straying (blue-dotted line) compared with a complete recovery with a very low stray rate of 0.01 (yellow-dashed line) in the no-barrier scenario. (iii) The inability of populations to recolonize eradicated patches when barriers are placed downstream of eradication zones.

programs for landscape genetic simulations focus on different processes and represent environmental heterogeneity in different ways. However, as landscape genetics simulation modelling platforms begin to proliferate, landscape genetics researchers have a unique opportunity to explore intermodel comparisons (similarly as with intermethod comparisons; e.g. Balkenhol *et al.* 2009a; Balkenhol, Waits & Dezzani 2009b), which allows further insights. It is quite possible that two different simulation studies will lead to complementary answers about a certain research question, thus allowing insights about the influences of model structure, parameters or other assumptions (e.g. Safner *et al.* 2011; Blair *et al.* 2012). For a general review of genetic simulation software, see Hoban (2014) or the Genetic Simulation Resources catalogue provided by the National Institute of Health (<https://popmodels.cancercontrol.cancer.gov/gsr/>).

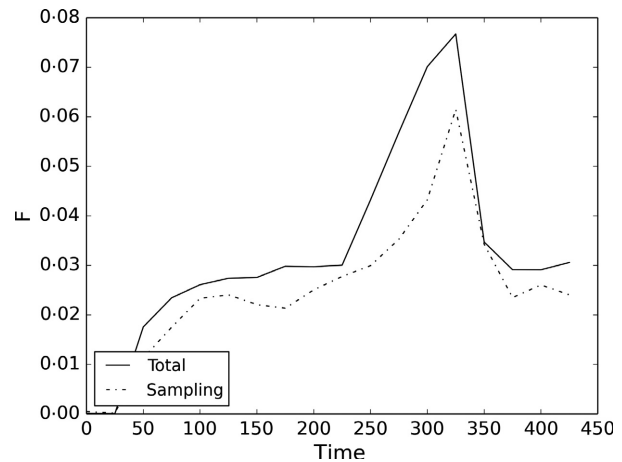


Fig. 3. Example #2 sampling around barrier detection and removal: Inbreeding coefficient (F) for Westslope Cutthroat Trout plotted across the 425-year simulation for all individuals (solid line), all individuals in the sampled 40 patches (dash-dotted line), and the captured individuals in the sampled 40 patches (dashed lines). Population dynamics burn-in period for the first 25 years, spatial genetic burn-in period using riverine distance from years 25 to 225, barrier insertion at year 225 and barrier removal at year 325.

Cost–Distance Meta-POPulation was developed from the program architecture of CDPOP (Landguth & Cushman 2010; Landguth, Cushman & Johnson 2012a; Landguth *et al.* 2012b), and a number of significant additions and modifications provide functionalities in CDMetaPOP that are not available in other existing landscape genetics platforms. Most noteworthy, CDMetaPOP couples genetic functionality with spatial demography that accounts for environmental heterogeneity as spatial and/or temporally varying quality of constraining factors at and between patch locations. CDMetaPOP provides a highly flexible approach of spatial extent and patch delineation, with the capacity to represent multiple attributes of landscapes that manifest (via interaction with important demographic processes) as the components driving population dynamics, while simultaneously controlling individual-based demographics and physiology. For example, individuals grow based on a spatially explicit environmental variable (e.g. temperature) and size-based relationships are associated with many processes (e.g. fecundity, maturation, movement rates). Genotype–environment associations can be specified and linked to specific processes, such as growth, maturation or stray rate. The resulting level of mortality, reflecting competition for limited resources, may be implemented via a density-dependent function with a hierarchical specification of patch- and class-structured carrying capacities. Finally, numerous movement processes between patches can be individually and explicitly controlled by different environmental variables, also considering directional influences (e.g. wind dispersal or one-way barriers) between patch locations.

Deciding whether or not simplifying assumptions that are implemented in a model are acceptable is a major component of conducting and evaluating simulation studies. While some abstraction of highly complex processes, such as growth and

density dependence, is necessary and treated in more detail in other programs (e.g. InSTREAM-Gen; Ayllon *et al.* 2016 or SimAdapt; Rebaudo *et al.* 2013), CDMetaPOP offers a uniquely flexible spatially explicit approach for understanding how complex demogenetic responses emerge from individuals interacting with variable and multiple environmental heterogeneities. Furthermore, CDMetaPOP incorporates a robust approach to stochasticity, via correlation structures and individual-based modelling, together with modules built to simulate environmental sampling and data collection. These features make this an ideal platform for exploring management strategies and generating expectations useful for evaluation of adaptive management approaches to complex, real-world natural resource management issues [e.g. suppression planning and barrier removal (e.g. Examples #1 and 2), or impacts of climate change or land use perturbations], as well as exploring eco-evolutionary theoretical studies tied to environmental gradients. However, we would like to remind end users to 'keep it simple' and that 'all models are wrong, but some are useful' (Box & Draper 1987). With simulation programs such as CDMetaPOP, it may seem easy to add another parameter or process to an existing model, but the potential for spurious results should not be overlooked. Therefore, we highly recommend conducting sensitivity, uncertainty and potentially, robustness analysis wherever possible.

Lastly, the program is built on a docking architecture that allows for ease of future development, which we envision will encompass much greater ecological realism, including individual variations in life-history strategies and behaviour, different modes of selection (e.g. natural vs. sexual) or trait coding (e.g. RangeShifter; Bocedi *et al.* 2014), alternative population models (e.g. Hassell 1975) and interactions among multiple species. We hope this open-source simulation framework stimulates additional modelling and applications to help identify populations vulnerable to environmental change, and to improve conservation and management of populations, species and ecosystems.

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Data accessibility

CDMetaPOP is a free, open-source program written in Python 2.7 object-oriented environment and is provided with a detailed user manual for installation instructions for most platforms, along with sample files and run instructions. CDMetaPOP has been debugged by testing all combinations of options and validated under Wright–Fisher assumptions. A user manual, example data sets and free downloads can be found at <https://github.com/ComputationalEcologyLab/CDMetaPOP>.

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